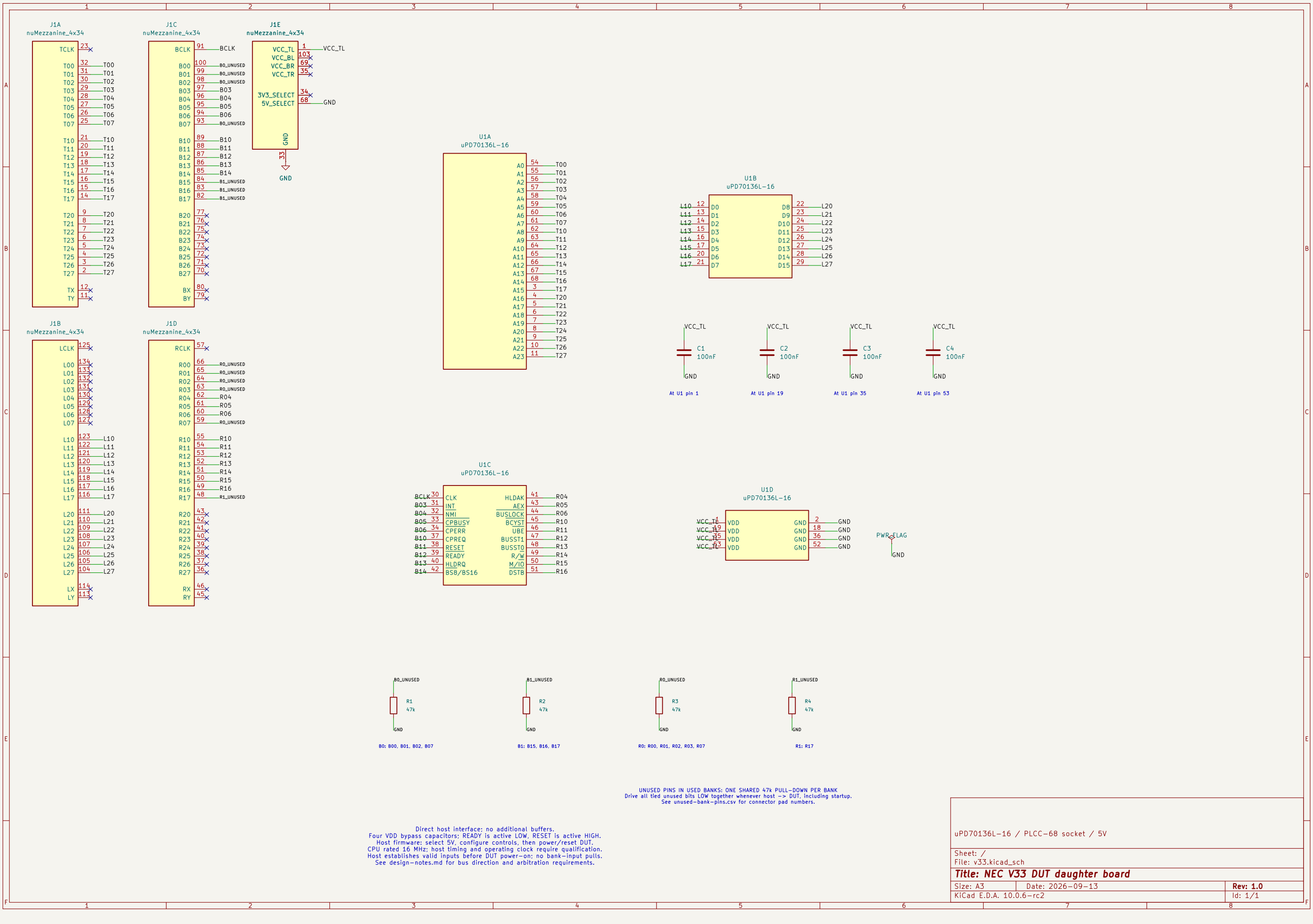

J1A
nuMezzanine_4x34
TCLK 23
T00 32 T00
T01 31 T01
T02 30 T02
T03 29 T03
T04 28 T04
T05 27 T05
T06 26 T06
T07 25 T07
T10 21 T10
T11 20 T11
T12 19 T12
T13 18 T13
T14 17 T14
T15 16 T15
T16 15 T16
T17 14 T17
T20 9 T20
T21 8 T21
T22 7 T22
T23 6 T23
T24 5 T24
T25 4 T25
T26 3 T26
T27 2 T27
TX 12
TY 11
J1C
nuMezzanine_4x34
BCLK 91 BCLK
B00 100 B0_UNUSED
B01 99 B0_UNUSED
B02 98 B0_UNUSED
B03 97 B03
B04 96 B04
B05 95 B05
B06 94 B06
B07 93 B0_UNUSED
B10 89 B10
B11 88 B11
B12 87 B12
B13 86 B13
B14 85 B14
B15 84 B1_UNUSED
B16 83 B1_UNUSED
B17 82 B1_UNUSED
B20 77
B21 76
B22 75
B23 74
B24 73
B25 72
B26 71
B27 70
BX 80
BY 79
J1E
nuMezzanine_4x34
VCC_TL 1 VCC_TL
VCC_BL 103
VCC_BR 69
VCC_TR 35
3V3_SELECT 34
5V_SELECT 68 GND
GND 33
GND
J1B
nuMezzanine_4x34
LCLK 125
L00 134
L01 133
L02 132
L03 131
L04 130
L05 129
L06 128
L07 127
L10 123 L10
L11 122 L11
L12 121 L12
L13 120 L13
L14 119 L14
L15 118 L15
L16 117 L16
L17 116 L17
L20 111 L20
L21 110 L21
L22 109 L22
L23 108 L23
L24 107 L24
L25 106 L25
L26 105 L26
L27 104 L27
LX 114
LY 113
J1D
nuMezzanine_4x34
RCLK 57
R00 66 R0_UNUSED
R01 65 R0_UNUSED
R02 64 R0_UNUSED
R03 63 R0_UNUSED
R04 62 R04
R05 61 R05
R06 60 R06
R07 59 R0_UNUSED
R10 55 R10
R11 54 R11
R12 53 R12
R13 52 R13
R14 51 R14
R15 50 R15
R16 49 R16
R17 48 R1_UNUSED
R20 43
R21 42
R22 41
R23 40
R24 39
R25 38
R26 37
R27 36
RX 46
RY 45
U1A
uPD70136L-16
A0 54 T00
A1 55 T01
A2 56 T02
A3 57 T03
A4 58 T04
A5 59 T05
A6 60 T06
A7 61 T07
A8 62 T10
A9 63 T11
A10 64 T12
A11 65 T13
A12 66 T14
A13 67 T15
A14 68 T16
A15 3 T17
A16 4 T20
A17 5 T21
A18 6 T22
A19 7 T23
A20 8 T24
A21 9 T25
A22 10 T26
A23 11 T27
U1B
uPD70136L-16
L10 12 D0
L11 13 D1
L12 14 D2
L13 15 D3
L14 16 D4
L15 17 D5
L16 20 D6
L17 21 D7
D8 22 L20
D9 23 L21
D10 24 L22
D11 25 L23
D12 26 L24
D13 27 L25
D14 28 L26
D15 29 L27
VCC_TL
C1
100nF
GND
At U1 pin 1
VCC_TL
C2
100nF
GND
At U1 pin 19
VCC_TL
C3
100nF
GND
At U1 pin 35
VCC_TL
C4
100nF
GND
At U1 pin 53
U1C
uPD70136L-16
BCLK 30 CLK
B03 31 INT
B04 32 NMI
B05 33 CPBUSY
B06 34 CPERR
B10 37 CPREQ
B11 38 RESET
B12 39 READY
B13 40 HLDRQ
B14 42 BS8/BS16
HLDAK 41 R04
AEX 43 R05
BUSLOCK 44 R06
BCYST 45 R10
UBE 46 R11
BUSST1 47 R12
BUSST0 48 R13
R/W 49 R14
M/IO 50 R15
DSTB 51 R16
U1D
uPD70136L-16
VCC_TL 4 VDD
VCC_TL 19 VDD
VCC_TL 35 VDD
VCC_TL 53 VDD
GND 2 GND
18 GND
36 GND
52 GND
PWR_FLAG
GND
B0_UNUSED
R1
47k
GND
B0: B00, B01, B02, B07
B1_UNUSED
R2
47k
GND
B1: B15, B16, B17
R0_UNUSED
R3
47k
GND
R0: R00, R01, R02, R03, R07
R1_UNUSED
R4
47k
GND
R1: R17
UNUSED PINS IN USED BANKS: ONE SHARED 47k PULL-DOWN PER BANK
Drive all tied unused bits LOW together whenever host -> DUT, including startup.
See unused-bank-pins.csv for connector pad numbers.
Direct host interface; no additional buffers.
Four VDD bypass capacitors; READY is active LOW, RESET is active HIGH.
Host firmware: select 5V, configure controls, then power/reset DUT.
CPU rated 16 MHz; host timing and operating clock require qualification.
Host establishes valid inputs before DUT power-on; no bank-input pulls.
See design-notes.md for bus direction and arbitration requirements.
uPD70136L-16 / PLCC-68 socket / 5V
Sheet: /
File: v33.kicad_sch
Title: NEC V33 DUT daughter board
Size: A3
Date: 2026-09-13
Rev: 1.0
KiCad E.D.A. 10.0.6-rc2
Id: 1/1